1. Competency #6: Identify and describe the mathematical aspects of a real-world problem situation, including identifying significant variables

> *"Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables"*

- Highly relevant: Students must extract core mathematical elements from the textual description:
  - Initial population: 10,000 adult penguins;
  - Reproduction rule: 5,000 pairs → 5,000 new chicks;
  - Mortality: 20% of the total individuals (adults + chicks) die by year-end.
- Key variables include: number of adults, number of chicks, total population, and death rate. Identifying these is the essential first step in modeling, fully aligning with Competency #6.

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2. Competency #7: Simplify or decompose a situation or problem to make it amenable to mathematical analysis

> *"Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis"*

- Highly relevant: Although the problem already provides simplifying assumptions (e.g., "one chick per pair," "uniform 20% mortality"), students must still break down the ecological process into computable steps:

  1. Start of year: 10,000 adults;
  2. Spring: +5,000 chicks → total = 15,000;
  3. End of year: 20% die → 80% survive → 0.8 × 15,000 = 12,000.
- This stepwise decomposition is precisely how real-world situations are made mathematically tractable, directly reflecting Competency #7.

3. Competency #12: Identify the constraints, assumptions, and simplifications in a mathematical model

> *"Identify the constraints, assumptions simplifications in a mathematical model"*

- Strongly relevant: The question explicitly lists three deliberate assumptions. Students must recognize that:
  - These are simplifications of reality (in nature, breeding success and mortality often vary by age or environment);
  - The model's output depends entirely on these premises;
  - Changing an assumption (e.g., only chicks experience 20% mortality) would yield a different result.
- While the task doesn't require critiquing the assumptions, correct solution hinges on accepting and applying them as given, which implicitly engages understanding of model structure—core to Competency #12.

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4. Competency #1: Select a mathematical description or representation that describes a problem

> *"Select a mathematical description or a representation that describes a problem"*

- Moderately relevant: Students must choose an appropriate mathematical approach—in this case, arithmetic operations with percentages, not differential equations or stochastic models.
- Although they aren't presented with multiple representation options, deciding that "addition and multiplication suffice" constitutes a strategic choice of mathematical representation, consistent with Competency #1.

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5. Competency #9: Translate a problem into a standard mathematical representation or algorithm

> *"Translate a problem into a standard mathematical representation or algorithm"*

- Moderately relevant: Students convert the narrative into a clear computational sequence:

$$\text{Final population} = (10{,}000 + 5{,}000) \times (1 - 0.2) = 15{,}000 \times 0.8 = 12{,}000$$

- This is a standard arithmetic algorithm, exemplifying the translation from natural language to mathematical operations.

6. Competency #5: Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations

> *"Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations"*

- Weakly relevant: Since the question covers only one year, it doesn't require identifying multi-year patterns (e.g., exponential decay). However, students still recognize the causal chain: reproduction → population increase → proportional mortality. This basic structural awareness aligns with Competency #5 at an introductory level.

1. 第6项：识别并描述现实问题中的数学要素，包括识别关键变量

> *"Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables"*

- 高度相关：学生必须从文字描述中提取核心数学元素：
  - 初始数量：10,000 成年企鹅；
  - 繁殖规则：5,000 对 → 5,000 只新雏鸟；
  - 死亡率：20% 的总个体（成年 + 雏鸟）在年底死亡。
- 关键变量包括：成年数、雏鸟数、总数量、死亡率。识别这些是建模的第一步，完全契合第6项。

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2. 第7项：简化或分解问题情境，使其适合数学分析

> *"Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis"*

- 高度相关：题干已提供简化假设（如“每对只生1只”“统一20%死亡率”），但学生仍需将复杂生态过程分解为可计算步骤：

  1. 年初：10,000 成年；
  2. 春天：新增 5,000 雏鸟 → 总数 = 15,000；
  3. 年底：20% 死亡 → 存活 80% × 15,000 = 12,000。
- 这种阶段性分解正是将现实问题转化为数学可处理形式的关键，直接体现第7项。

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3. 第12项：识别数学模型中的约束、假设与简化

> *"Identify the constraints, assumptions simplifications in a mathematical model"*

- 强相关：题目明确列出三项人为假设（每对生1雏、20%死亡率、年初10,000只），学生需意识到：
  - 这些假设是对现实的简化（现实中繁殖成功率、死亡率可能因年龄而异）；
  - 模型结果依赖于这些前提；
  - 若改变假设（如死亡率仅适用于雏鸟），结果会不同。
- 虽然题目未要求批判性评价，但正确解题的前提是接受并应用这些给定假设，隐含了对模型结构的理解。

4. 第1项：选择能描述问题的数学表述或表示方式

> *"Select a mathematical description or a representation that describes a problem"*

- 中等相关：学生需选择合适的数学工具来表达过程——此处是算术序列与百分比运算，而非微分方程或概率模型。
- 虽然未显式“选择”多种表示，但判断“用加法+乘法即可解决”本身是一种表示策略的决策，符合第1项精神。

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5. 第9项：将问题转化为标准数学表示或算法

> *"Translate a problem into a standard mathematical representation or algorithm"*

- 中等相关：学生将文字描述转化为以下计算流程：

$$\text{Final} = (10{,}000 + 5{,}000) \times (1 - 0.2) = 15{,}000 \times 0.8 = 12{,}000$$

- 这是一个标准化的算术算法，体现了从自然语言到数学操作的翻译过程。

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6. 第5项：识别问题或情境中的数学结构（如规律、关系、模式）

> *"Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations"*

- 弱相关：本题仅涉及单一年份，未要求发现跨年度模式（如指数衰减/增长）。但学生仍需识别“繁殖→总数增加→按比例死亡”这一因果链结构，属于基础层面的结构识别。